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SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF COMPUTER ENGINEERING

TOPIC: Multi-Source Energy-Harvesting Autonomous Sensor Node.

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LITERATURE REVIEW

1. Introduction

The rapid growth of Wireless Sensor Networks (WSNs) and Internet of Things (IoT) applications in areas such as environmental monitoring, smart agriculture, industrial automation, and infrastructure surveillance has intensified the demand for long-term, maintenance-free sensor operation. Traditionally, most sensor nodes rely on batteries as their primary power source. While batteries provide predictable energy supply, they introduce major limitations, including limited operational lifetime, frequent maintenance, high replacement costs, and environmental concerns related to battery disposal. These challenges are particularly significant in remote, hazardous, or large-scale deployments where physical access to sensor nodes is difficult or costly.

Energy harvesting has emerged as a promising alternative to battery-powered sensor nodes by enabling devices to extract energy from ambient sources such as solar radiation, magnetic fields, and radio frequency (RF) signals. However, many early energy harvesting systems relied on a single ambient source, making them vulnerable to environmental variability and intermittency. As a result, such systems often fail to guarantee reliable and continuous operation. This has motivated research into multi-source energy harvesting combined with efficient power management and adaptive duty-cycling mechanisms. This chapter reviews prior academic studies and existing systems to identify their strengths, limitations, and unresolved challenges, thereby justifying the need for the proposed multi-source energy-harvesting autonomous sensor node.

2. Related Work (Academic Prior Studies)

Early research in ambient energy harvesting demonstrated the feasibility of powering low-power electronic devices using scavenged environmental energy. Foundational studies established that micro-watt to milli-watt power levels could support basic sensing, processing, and communication tasks if energy consumption was carefully managed. This led to the concept of energy-neutral operation, where a sensor node dynamically adapts its workload to match harvested energy availability.

Because all ambient energy sources are inherently intermittent, energy storage plays a critical role in autonomous sensor systems. While rechargeable batteries provide high energy density, their limited cycle life and aging effects reduce suitability for long-term maintenance-free deployment. Super capacitors have therefore gained increasing attention due to their fast charging capability, long cycle life, and compatibility with battery-less energy harvesting architectures.

2.2 Ambient Energy Harvesting Techniques

Research by Paradiso and Sterner demonstrated the feasibility of powering low-power electronics using ambient energy. Solar, piezoelectric, and thermoelectric harvesting techniques have been widely explored, though each suffers from environmental dependency namely.

2.2.1 Solar Energy Harvesting

Solar energy harvesting has been the most extensively studied technique due to its relatively high power density and technological maturity. Photovoltaic cells can generate significant power under favorable lighting conditions; however, their output is highly dependent on irradiance, temperature, and panel orientation. To address these variations, Maximum Power Point Tracking (MPPT) techniques are widely employed to maximize energy extraction. Despite these improvements, solar harvesting remains intermittent and unreliable during nighttime or poor weather conditions.

Solar energy harvesting is one of the most extensively researched and practically deployed ambient energy harvesting techniques due to its comparatively high energy density and technological maturity. It employs photovoltaic (PV) cells to convert incident solar radiation directly into electrical energy through the photovoltaic effect **(Paradiso & Starner, 2005)**. Because solar radiation provides higher power levels than most other ambient sources, PV harvesting is frequently adopted as the primary energy source for outdoor and semi-outdoor wireless sensor systems **(Roundy, Wright, & Rabaey, 2004).**

Experimental and analytical studies indicate that PV-based harvesters can produce output power ranging from a few microwatts under indoor or low-light conditions to several milli-watts under direct sunlight, depending on illumination intensity, panel surface area, material efficiency, and temperature **(Roundy et al., 2004)**. However, the electrical characteristics of PV cells are highly nonlinear. Variations in irradiance and temperature significantly affect output voltage and current, making direct coupling to electronic loads inefficient and potentially unstable.

To mitigate these effects, Maximum Power Point Tracking (MPPT) techniques are widely employed to dynamically adjust the operating point of the PV cell and maximize harvested energy under changing environmental conditions **(Esram & Chapman, 2007).** Despite these advancements, solar energy harvesting remains inherently intermittent, as it depends on daylight cycles and weather conditions, limiting its reliability as a standalone energy source **(Paradiso & Starner, 2005).**

2.2.2 Magnetic Field Energy Harvesting

Magnetic field energy harvesting exploits alternating magnetic fields generated by electrical infrastructure such as power transmission lines, transformers, motors, and industrial machinery **(Beeby & White, 2010)**. This technique typically utilizes inductive coils to convert time-varying magnetic flux into electrical energy based on Faraday's law of electromagnetic induction **(Roundy, Wright, & Rabaey, 2004)**.

Magnetic energy harvesting is particularly advantageous in indoor and industrial environments where strong magnetic fields are continuously present. Unlike solar harvesting, it is independent of lighting conditions and can provide energy as long as the source equipment remains operational **(Beeby & White, 2010)**. This makes it attractive for environments where light availability is limited or inconsistent.

However, the harvested power levels are generally low and strongly dependent on coil geometry, distance from the magnetic source, operating frequency, and field strength **(Roundy, Wright, & Rabaey, 2004)**. As a result, magnetic harvesting alone is insufficient to sustain continuous sensing and wireless communication. Research indicates that it is most effective as a supplementary energy source when combined with efficient rectification, voltage boosting, and intelligent energy management circuits **(Beeby & White, 2010)**.

2.2.3 Radio Frequency (RF) Energy Harvesting

Radio Frequency (RF) energy harvesting captures energy from ambient electromagnetic waves emitted by radio transmitters, television stations, cellular base stations, and Wi-Fi access points **(Paradiso & Starner, 2005).** A typical RF harvesting system consists of an antenna, impedance matching network, rectifier, and energy storage element, commonly referred to as ***antenna***. **(Sample, Yeager, Powledge, Mamishew, & Smith, 2008).**

One advantage of RF harvesting is its availability in both indoor and outdoor environments, particularly in urban areas where RF signals are continuously present. However, the ambient RF power density is extremely low, often in the Nano-watt to microwatt range, making RF harvesting unsuitable as a primary energy source for most sensor applications **(Roundy, Wright, & Rabaey, 2004)**.

Despite these limitations, RF harvesting plays an important role in battery-less and ultra-low-power systems. It is particularly effective for standby operation, periodic wake-up events, and trickle charging of energy storage components when combined with high-efficiency rectification and impedance matching **(Sample, Yeager, Powledge, Mamishew, & Smith, 2008).** Consequently, RF harvesting is most effective when integrated into multi-source energy harvesting architectures.

2.2.4 Energy Storage Technologies for Energy Harvesting

Energy storage is a fundamental component of energy harvesting systems due to the intermittent and unpredictable nature of ambient energy sources **(Priya & Inman, 2009)**. The primary function of storage is to buffer harvested energy and provide a stable and continuous power supply to electronic loads during periods of insufficient energy availability.

Rechargeable batteries, such as lithium-ion and lithium-polymer types, offer high energy density and support long-duration operation. However, they suffer from limited charge discharge cycles, aging effects, leakage, and environmental concerns, which reduce their suitability for long-term, maintenance-free deployments **(Beeby & White, 2010)**.

Super capacitors, also known as ultra-capacitors, provide high power density, rapid charging capability, and extremely long cycle life, often exceeding millions of cycles without significant degradation **(Priya & Inman, 2009)**. These characteristics make them particularly suitable for battery-less energy harvesting systems. Consequently, super capacitors and thin-film storage devices are increasingly preferred, with power management integrated circuits (PMICs) playing a critical role in voltage regulation, energy protection, and system stability **(Beeby & White, 2010)**.

2.2.5 Kinetic and Aero-elastic Energy Harvesting:

Recent research has moved beyond simple vibration to exploit aero-elastic flutter for industrial and aerospace applications. By using piezoelectric materials (such as PZT-5A) attached to wing-like structures, researchers have demonstrated the ability to generate up to **10.8mW** of power from airflow far exceeding traditional vibration harvesters **(Erturk, Vieira, De Marqui, & Inman, 2010)**. This provides a critical solution for monitoring remote infrastructure like wind turbines or aircraft sensors where solar and thermal gradients are inconsistent.

2.2.6 Cognitive Radio and Spectrum Efficiency:

A major bottleneck in Energy Harvesting WSNs (EH-WSNs) is the energy wasted during "spectrum sensing" and data collisions. Modern studies integrate Cognitive Radio (CR) to allow energy-neutral nodes to automatically detect and utilize unused spectrum bands. This "Energy-Aware Spectrum Management" ensures that nodes only transmit when a clear channel is found, preventing the wasted energy of re-transmissions a vital strategy for maintaining the stability of the energy buffer in low-power environments **(Akan, Karli, & Ergul, 2009)**.

2.2.7 Check pointing for Intermittent Computing:

Since ambient energy is often highly transient, a significant body of recent work focuses on **state-check pointing mechanisms**. Unlike traditional sensors that restart their software upon power failure, state-of-the-art systems save the program state to non-volatile memory (like FRAM) once energy drops below a critical threshold. This allows the node to resume its specific task exactly where it left off once energy returns, preventing the "reboot-loop" that often renders battery-less devices ineffective during fluctuating power conditions **(Ransford, Lucia, & Levis, 2011)**.

2.2.8 Emerging Indoor Photovoltaics:

While silicon-based solar is mature, current research in **Perovskite Solar Cells (PSCs)** and **Organic Photovoltaics (OPVs)** is revolutionizing indoor IoT. These materials are optimized specifically for the spectrum of LED and fluorescent lighting, achieving higher efficiency in low-lux environments than traditional outdoor panels. These thin-film, flexible cells allow for the integration of energy harvesting directly into wearable medical sensors and retail "smart labels" that must operate under constant but low-intensity indoor light **(IEEE, 2025)**.

2.2.9 Hybrid Energy Management and Machine Learning:

The most recent advancements involve **Hybrid Energy Harvesting**, which simultaneously scavenges from solar, thermal, and RF sources. To manage these complex inputs, researchers are employing **TinyML (on-device Machine Learning)** to predict future energy availability. These intelligent systems dynamically adjust the node's duty cycle increasing sensing frequency during "energy surplus" and entering ultra-low-power sleep during "energy droughts "to achieve a 66% reduction in discovery delay and significantly prolong the network's operational life **(Warden & Situnayake, 2020)**.

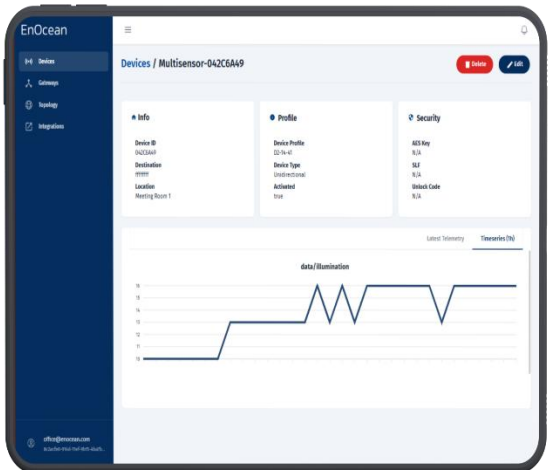
2.2.10 Computational RFID and Battery-Free Sensing (The WISP Project):

A unique branch of energy harvesting research focuses on **Computational RFID**, exemplified by the **Wireless Identification and Sensing Platform (WISP)** developed initially by Intel Research and the University of Washington **(Sample, Yeager, Powledge, Mamishew, & Smith, 2008)**. Unlike traditional RFID tags that only transmit a static ID, WISP is a fully programmable, battery-free sensing and computing platform. It harvests energy from the radio frequency (RF) energy emitted by standard UHF RFID readers to power a 16-bit microcontroller and a variety of sensors (accelerometers, temperature, and strain gauges). The WISP project is significant because it introduced the concept of **intermittent computing**. Research using this platform addresses the "reboot-loop" problem where a device starts a task, runs out of power, and loses its progress. WISP-related studies have pioneered software techniques that allow a sensor to save its state into non-volatile memory (like FRAM) just before power failure and resume immediately upon the next energy harvest. This work serves as the academic foundation for modern battery-less IoT devices that must perform complex logic in highly transient energy environments **(Sample, Yeager, Powledge, Mamishew, & Smith, 2008)**.

3 Previous Systems (or Similar Applications)

This section reviews existing systems and applications that are functionally related to the proposed **multi-source energy-harvesting, battery-less autonomous sensor node**. The reviewed systems provide energy harvesting, sensing, monitoring, or management capabilities similar to the proposed solution.

3.1 EnOcean Device Management Platform







- **System name:** EnOcean Device Management Platform
- **URL:** https://www.enocean.com/en/products/self-powered-sensors
- **Platform:** Windows, Linux, macOS via browser
- **Trial license:** Demo kits (vendor-controlled)

Description:

This platform was developed to configure and monitor EnOcean self-powered sensors used in building automation. It targets building managers and system integrators seeking maintenance-free sensor deployments. While reliable, it employs conservative duty cycles and single-source harvesting, limiting adaptability compared to the proposed system. EnOcean self-powered sensors are commercially deployed devices that harvest energy from indoor light or mechanical motion to enable battery-less operation. They target building automation users but operate with conservative duty cycles **(EnOcean GmbH, 2026)**.

Limitations

The EnOcean platform is optimized primarily for ultra-low-power building automation scenarios, where environmental conditions are relatively predictable and energy harvesting typically relies on indoor solar or kinetic sources. While this enables maintenance-free deployments, the architecture assumes constrained workloads and conservative duty cycles to preserve energy stability. The system does not support dynamic integration of multiple heterogeneous energy sources, nor does it provide real-time energy estimation at the firmware level.

Unresolved Problems

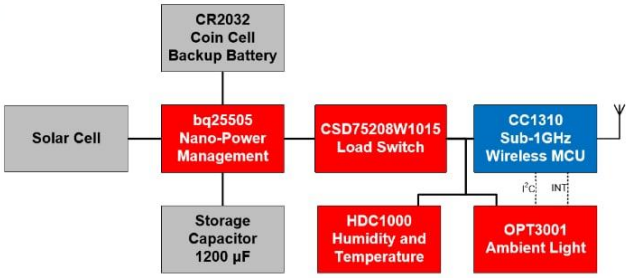
The platform does not address the coordination of diverse harvesting modalities under fluctuating environmental conditions. It lacks predictive energy modeling, adaptive communication scheduling, and workload prioritization based on instantaneous storage levels. Furthermore, scalability beyond building automation contexts for example, industrial outdoor environments remain insufficiently demonstrated.

Need for Current Research

Future research must extend beyond single-source reliability toward multi-source adaptive energy orchestration. A unified architecture that integrates heterogeneous energy inputs and dynamically modulates sensing and transmission behavior is required. The proposed multi-source autonomous sensor node addresses these limitations by implementing real-time energy-aware duty cycling and diversified harvesting inputs.

3.2 Texas Instruments Energy Harvesting Evaluation GUI (BQ25570)







- **System name:** TI Energy Harvesting Evaluation GUI (part of the BQ25570EVM-206 kit)
- **URL:** https://www.ti.com/product/BQ25570
- **Platform:** Windows
- **Trial license:** Free evaluation software

Description:

Developed to support evaluation of TI energy-harvesting PMICs, this application visualizes voltage levels and MPPT behavior. It targets embedded system designers and researchers. The software focuses on hardware characterization, such as verifying cold-start voltages and continuous harvesting efficiency down to 100 mV, but it lacks autonomous sensing and system-level intelligence.

Limitations

The TI BQ25570 platform is fundamentally a hardware evaluation environment intended to characterize ultra-low-power boost converters and MPPT behavior. While it excels in demonstrating voltage regulation efficiency and cold-start performance, it does not extend into system-level integration. There is no built-in sensing logic, communication stack, or adaptive workload management capability.

Unresolved Problems

Although the tool optimizes energy extraction efficiency, it does not solve how harvested energy should be allocated across sensing, processing, and communication tasks. The absence of cross-layer integration between hardware and firmware such as warning microcontrollers of pending power loss via the programmable threshold flags, remains a significant gap for many developers. Additionally, multi-source coordination strategies are not addressed within this single-chip framework.

Need for Current Research

Research must transition from component-level efficiency optimization toward holistic autonomous system design. Integrating PMIC regulation with firmware-level energy budgeting and adaptive workload management is necessary for practical battery-less operation. The proposed project expands beyond hardware characterization toward an intelligent energy-driven sensing architecture that utilizes the BQ25570's power-good flags and nano-power buck converter for dynamic system scaling.

3.3 Power cast RF Energy Harvesting Monitoring System







- **System name:** Power cast Energy Monitoring Software
- **URL:** https://www.powercastco.com/technologies/rf-power
- **Platform:** Web-based / Windows
- **Trial license:** Hardware evaluation kits (time-limited)

Description:

This system monitors RF-powered IoT devices in wireless power environments. It targets engineers deploying RF-powered sensors. Dependence on dedicated RF transmitters and lack of hybrid harvesting limit its flexibility. Their technology uses a "Power caster" transmitter that broadcasts radio waves (usually in the 915MHz ISM band). The receiving end uses a high-efficiency **rectenna**

(rectifying antenna) to convert those radio waves back into DC electricity. This is particularly useful in industrial settings where sensors are buried in walls or located in rotating machinery were changing batteries or using solar is impossible **(Powercast Corporation, 2025)**.

Limitations

Power cast systems demonstrate effective RF-based wireless power transfer; however, practical deployment typically requires dedicated RF transmitters to achieve usable power levels. Ambient RF energy density in uncontrolled environments remains insufficient for sustained sensing and wireless communication.

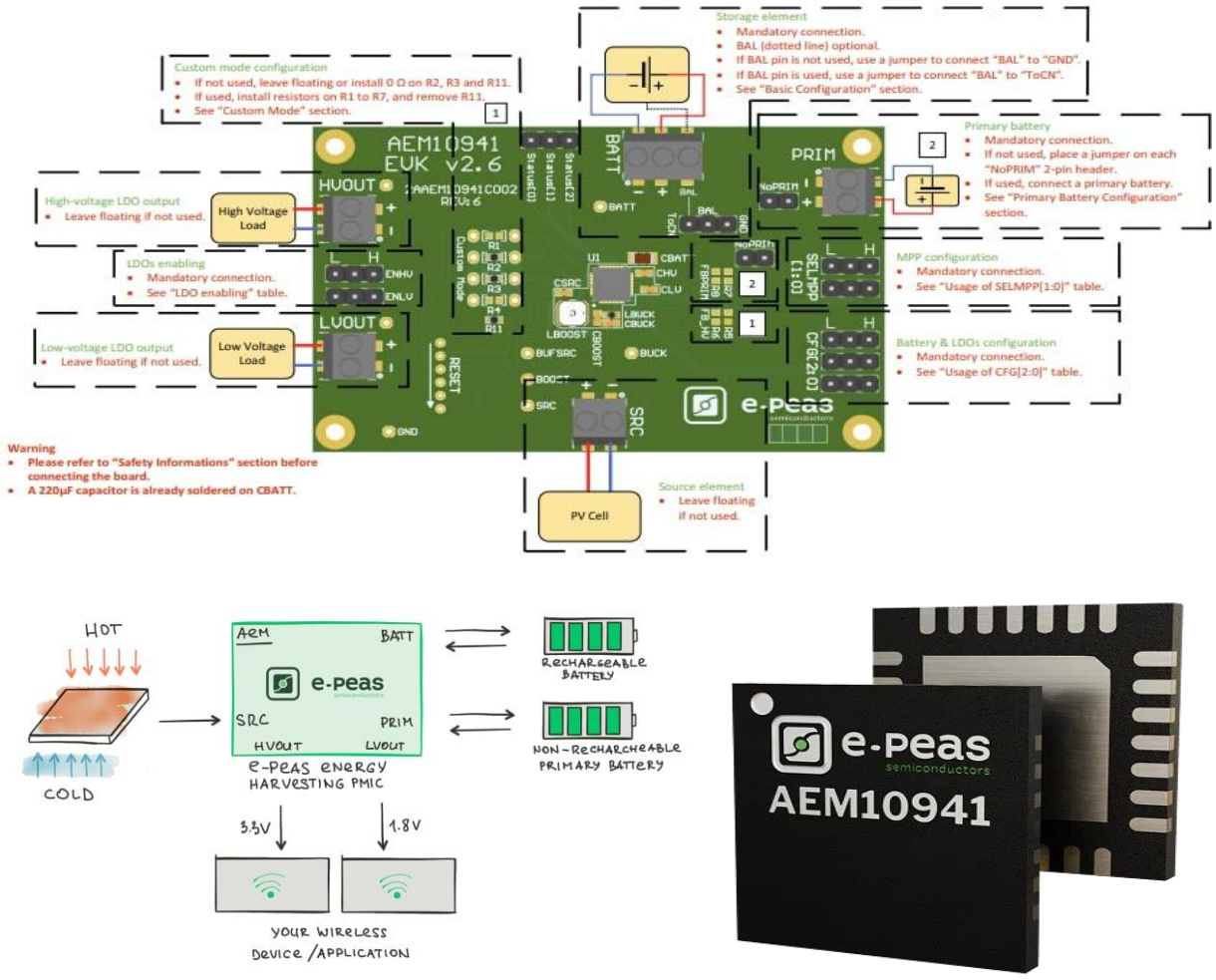
Unresolved Problems

Reliance on infrastructure-based RF transmitters limits scalability and cost-efficiency. Additionally, the architecture does not integrate alternative energy sources to mitigate RF intermittency or low ambient power levels. Energy storage management under fluctuating RF conditions is also minimally addressed.

Need for Current Research

Hybrid architectures that treat RF harvesting as a supplementary source within a broader multi-source ecosystem are required. Research must investigate coordinated harvesting where RF energy complements solar and magnetic inputs to improve resilience. The proposed system directly responds to this need by incorporating RF harvesting as one component of a diversified energy strategy.

3.4 E-peas Energy Harvesting Demonstration Software (AEM10941)



- **System name:** e-peas Energy Harvesting Demo Software
- **URL:** https://e-peas.com/product/aem10941
- **Platform:** Windows
- **Trial license:** Evaluation kits

Description:

Designed to demonstrate photovoltaic harvesting with MPPT and super capacitor storage, this software targets hardware engineers. It is hardware-centric and does not implement predictive or adaptive task scheduling. The **AEM10941** is often the top choice for photovoltaic (PV) harvesting because of its ultra-low power footprint. It can start harvesting energy when the input voltage is as low as 380mV, which is critical for indoor solar panels that may only produce a tiny amount of

voltage. It manages the entire energy chain: it extracts power from the PV cell, stores it in a battery or super capacitor, and provides two regulated voltages to power an application's MCU and radio (e-peas, 2024).

Limitations

The e-peas AEM10941 platform provides high-efficiency photovoltaic harvesting with MPPT and super capacitor compatibility. However, it remains hardware-centric, focusing on energy conversion performance rather than autonomous workload adaptation.

Unresolved Problems

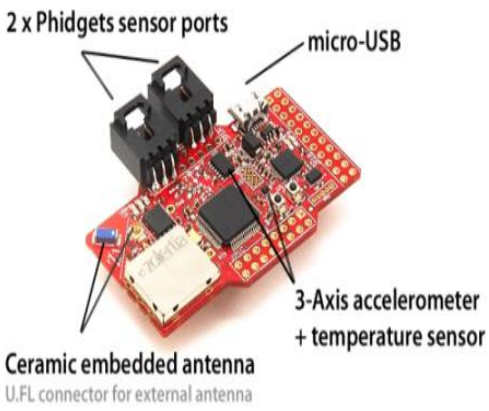
There is limited integration between harvested energy metrics and real-time system behavior adjustment. The system does not incorporate predictive duty cycling or adaptive communication strategies based on stored energy thresholds.

Need for Current Research

Future designs must merge efficient PMIC technology with intelligent firmware capable of adjusting sensing intervals, transmission frequency, and sleep states dynamically. The proposed system advances this direction by embedding adaptive duty-cycling algorithms into the harvesting architecture.

3.5 Zolertia IoT Energy-Harvesting Node (Zolertia IPv6/6LoWPAN Ubidots client)





- **System name:** Zolertia IPv6/6LoWPAN Ubidots client
- **URL:** https://www.hackster.io/4348/zolertia-ipv6-6lowpan-ubidots-bcde84
- **Platform:** Web-based
- **Trial license:** Open-source / research use

Description:

This dashboard supports monitoring of low-power IoT nodes in research deployments. Target users include researchers and universities. Most nodes remain battery-assisted, limiting full autonomy. Zolertia platforms, such as the Z1 mote used in the referenced IPv6/6LoWPAN project, are engineered for the Industrial Internet of Things (IoT). They employ a Texas Instruments CC2538 SoC with dual-radio support for flexible 2.4GHz and sub-GHz communication, forming resilient mesh networks. A core design philosophy is its "harvesting-ready" architecture: the hardware includes dedicated headers and low-quiescent-current components, allowing for the straightforward integration of external energy-harvesting modules for example solar, wind, or thermal **(Lignan, Antonio, 2015)**.

Limitations

Although Zolertia platforms support low-power IoT experimentation, many deployments remain battery-assisted. Energy harvesting capabilities are not consistently integrated into adaptive scheduling frameworks.

Unresolved Problems

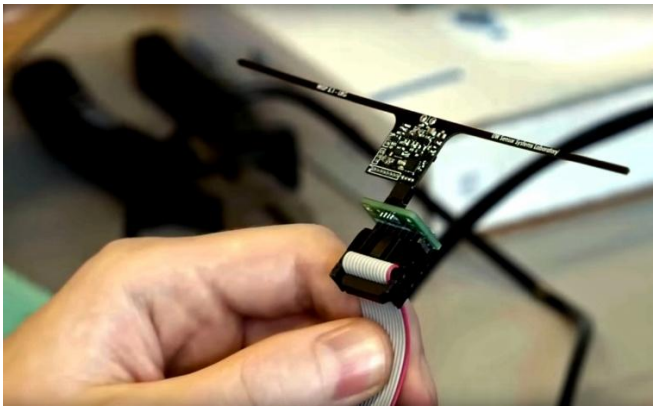
Energy-neutral operation without battery backup remains insufficiently demonstrated. Real-time energy estimation and dynamic workload prioritization are not fully integrated into the monitoring architecture.

Need for Current Research

A truly battery-less system requires multi-source harvesting combined with firmware-level decision-making that eliminates fallback battery dependency. The proposed research addresses this by designing a fully autonomous node without reliance on auxiliary battery support.

3.6 Wireless Identification and Sensing Platform (WISP) Tools





- **System name:** WISP Monitoring and Debug Tools
- **URL:** https://sites.google.com/uw.edu/wisp-wiki/home
- **Platform:** Windows, Linux and MacOS (PC-based research tools)
- **Trial license:** Open research platform

Description:

Developed to demonstrate battery-free RF-powered sensing, WISP targets academic researchers. Operation is constrained by reader proximity and intermittent power.

Limitations

The **Wireless Identification and Sensing Platform (WISP)** is an open source and fully hackable battery-free platform for experimentation with low-power sensing, computation, and communication **(Smith, 2021)**. WISP technology depends heavily on RFID reader proximity for energy harvesting. The intermittent nature of RF supply causes unstable sensing cycles and limits operational continuity.

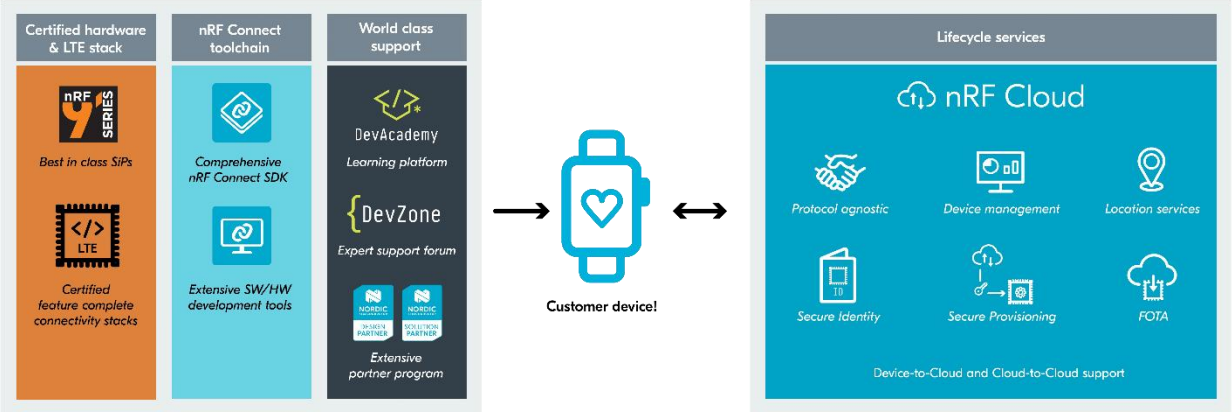
Unresolved Problems

Continuous environmental monitoring is impractical due to inconsistent power availability. The system lacks integration with other ambient sources to stabilize operation under varying environmental conditions.

Need for Current Research

A multi-source architecture that supplements RF harvesting with solar and magnetic energy can mitigate intermittency and improve reliability. The proposed system directly expands on WISP's concept by removing single-source dependency.

3.7 Nordic Semiconductor BLE Energy Profiling Tools



- **System name:** Nordic Power Profiler
- **URL:** https://www.nordicsemi.com/Products/Wireless/Low-power-cellular-IoT
- **Platform:** Windows
- **Trial license:** Free with hardware

Description:

The Nordic Power Profiler is a dedicated analysis tool designed for embedded developers to measure and evaluate the power consumption of Bluetooth Low Energy (BLE) devices. It focuses on quantifying how much energy specific operations—especially wireless communications—use in real application scenarios. While the tool offers detailed insight into where and how energy is consumed, it does not include features for energy harvesting or autonomous power management; rather, it highlights that communication operations often dominate overall energy use.

Limitations

The Nordic Power Profiler functions only as a measurement and profiling instrument; it does not provide any mechanisms for harvesting energy from the environment.

It does not include adaptive workload control or dynamic power management strategies that react to available energy.

Unresolved Problems

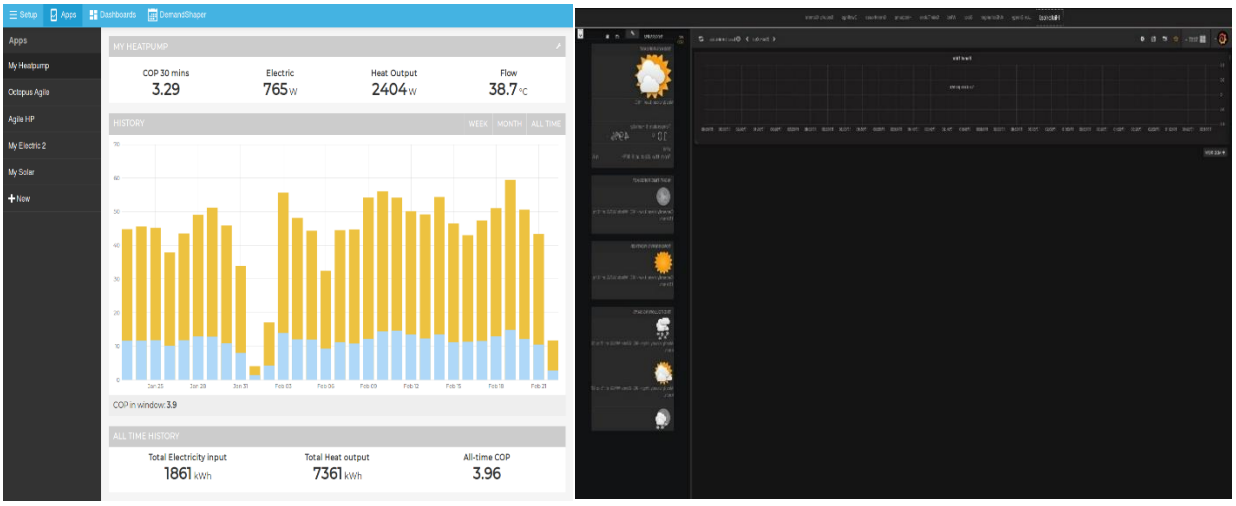
Although the Profiler can identify high-consumption elements such as BLE radio activity it does not offer integrated methods to *respond* to those findings under constrained energy conditions. For example:

- It cannot automatically reduce transmission frequency when battery levels are low of which an multi-energy sources would sustain the system.
- It lacks native support for shaping device workflows based on consumption patterns

Need for Current Research

Research must integrate consumption profiling insights with adaptive firmware strategies to dynamically reduce communication overhead when harvested energy is insufficient. The proposed system incorporates energy-aware scheduling mechanisms inspired by such profiling insights.

3.8 Open Energy Monitor IoT Monitoring Platform



- **System name:** Open Energy Monitor
- **URL:** https://openenergymonitor.org
- **Platform:** Web-based
- **Trial license:** Open-source

Description:

Originally developed for energy monitoring, this platform supports sensor data visualization. It targets hobbyists and researchers. Battery-less harvesting is not native, limiting suitability for autonomous sensing.

Limitations

The platform focuses on energy data visualization rather than autonomous energy management. It assumes stable and continuous power supply.

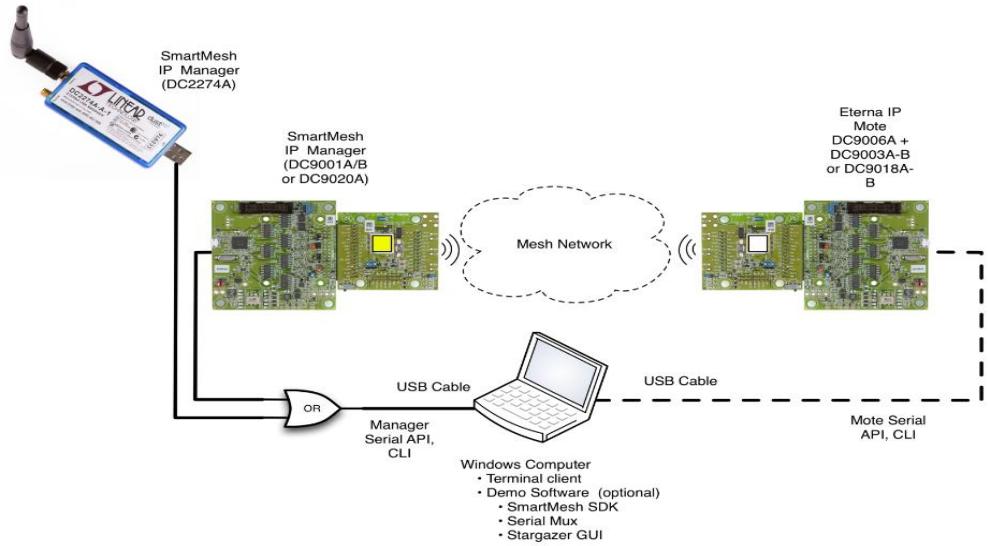
Unresolved Problems

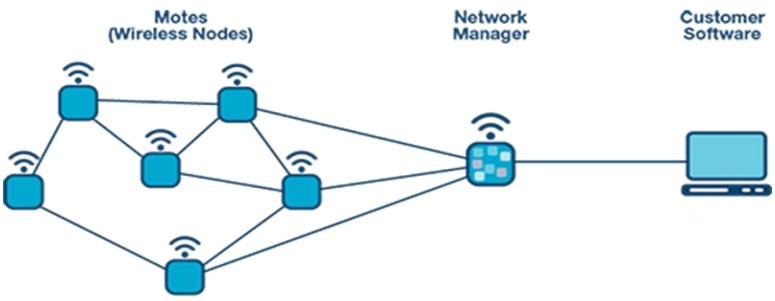
There is no mechanism to adapt sensing frequency or communication intervals based on available harvested energy.

Need for Current Research

Future IoT platforms must integrate energy-awareness at the application layer. The proposed research contributes by embedding energy-level-based workload modulation within the sensing node itself.

3.9 Smart Mesh IP Power Monitoring Tools





- **System name:** Smart Mesh Network Manager
- **URL:** https://www.analog.com/media/en/technical-documentation/user-guides/smartmesh_ip_tools_guide.pdf
- **Platform:** Web-based / PC
- **Trial license:** Evaluation-based

Description:

This system manages ultra-low-power wireless sensor networks. It targets industrial IoT users. Nodes typically rely on batteries rather than ambient harvesting **(Analog Devices, 2021)**.

Limitations

Existing solutions often lack integrated Maximum Power Point Tracking (MPPT) at the network protocol level. While chips like the e-peas AEM10941 support hardware-level MPPT ratio configuration for PV cells, this logic is not typically communicated to the mesh routing layer. Additionally, batteries remain finite and create maintenance cycles that energy harvesting aims to eliminate.

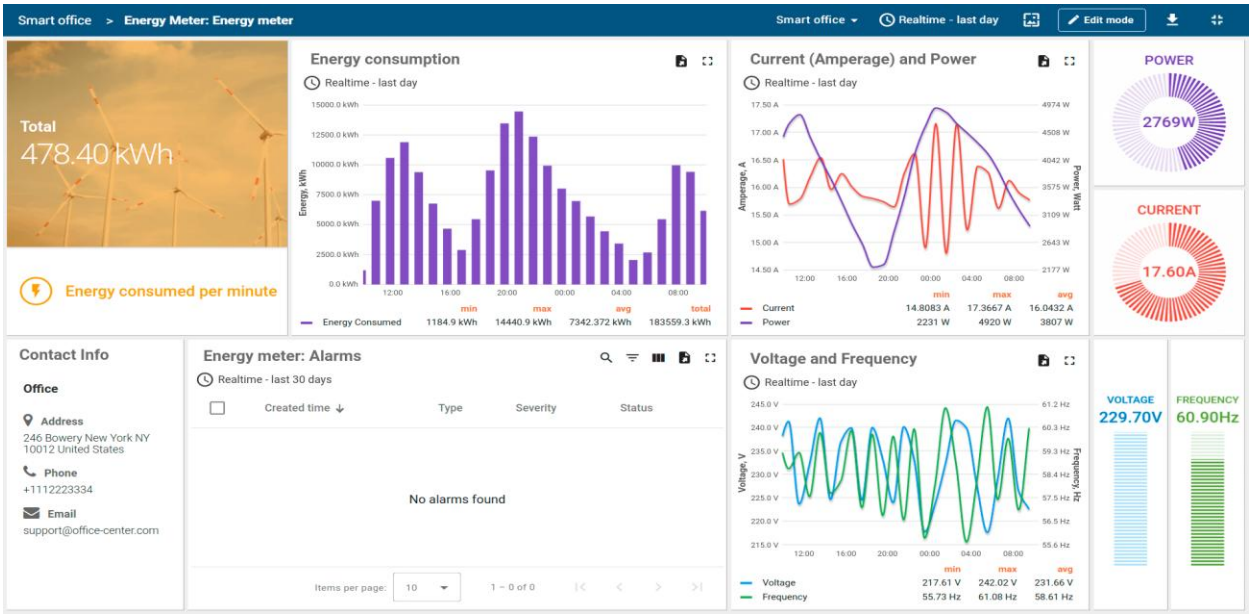
Unresolved Problems

Energy-neutral mesh operation under fluctuating harvested power remains insufficiently explored. A major challenge is maintaining a continuous power rail for high-demand "burst" operations, such as wireless data transmission, when harvested energy is minimal. Ensuring **cold-start capability** from extremely low voltages remains a bottleneck for starting up a mesh network after a period of total darkness or thermal inactivity.

Need for Current Research

There is a critical need to research **adaptive monitoring** that adjusts reporting frequency based on the current state of a node's energy buffer for example, super capacitors or thin-film batteries). Research into **autonomous power management** like that found in EnOcean's multi-sensors which combine solar harvesting with temperature and humidity sensing is essential for creating a "digital twin" of industrial environments that requires zero maintenance.

3.10 Things Board IoT Monitoring Platform



- **System name:** Things Board
- **URL:** https://thingsboard.io
- **Platform:** Web-based
- **Trial license:** Community Edition (free)

Description:

Things Board is a generic IoT monitoring platform used to visualize sensor data. It targets IoT developers and enterprises. Energy harvesting awareness and adaptive duty cycling are not natively supported.

Limitations

Things Board functions as a telemetry visualization and rule engine platform without intrinsic energy-awareness mechanisms.

Unresolved Problems

The platform assumes continuous device uptime and stable connectivity, which is incompatible with intermittent energy harvesting conditions.

Need for Current Research

IoT management platforms must evolve toward energy-adaptive frameworks that support variable communication intervals. The proposed research demonstrates how energy-aware node behavior can complement such platforms.

4. Lessons Learnt from the Review

The literature as reviewed or rather highlighted several key lessons on the systems. First, reliance on a single ambient energy source is insufficient for reliable long-term operation due to environmental variability. Multi-source energy harvesting significantly improves robustness by compensating for the absence or degradation of individual sources.

Second, hardware-focused evaluation tools alone are inadequate for autonomous operation without intelligent firmware capable of energy estimation and adaptive control.

Third, RF energy harvesting is most effective as a supplementary source rather than a primary power supply.

Fourth, super capacitor-based storage offers clear advantages for battery-less systems due to long cycle life and fast charging.

Finally, adaptive duty cycling is essential for aligning energy consumption with harvested energy and preventing system brownout.

5. A Critique of the Review (Gaps, Inconsistencies, Omissions)

Despite extensive research and numerous existing systems, several gaps remain unresolved. Many prior works focus on isolated components. Such as harvesters, power management circuits, or monitoring dashboards rather than integrated autonomous sensor systems. Single-source dependence remains common, even though environmental variability is well known. Furthermore, most IoT platforms lack energy-awareness and assume stable power availability, making them unsuitable for battery-less operation.

Another major omission is the limited integration of energy harvesting with adaptive firmware-level decision-making. While MPPT and storage are widely addressed, fewer systems incorporate intelligent duty cycling based on real-time energy availability. Additionally, many studies evaluate performance under laboratory conditions without addressing long-term deployment variability. These gaps collectively justify the need for a unified multi-source energy-harvesting system that integrates power management, energy storage, and adaptive control to achieve reliable autonomous operation.

6. Conclusion

This literature review has shown that although energy harvesting technologies are well established, achieving fully autonomous battery-less sensor nodes remains an open challenge. Existing studies and systems provide valuable insights but often address only partial aspects of the problem. Integrating multiple ambient energy sources with efficient power management, appropriate energy storage, and adaptive duty-cycling mechanisms offers a viable pathway toward sustainable, maintenance-free IoT deployments. The identified limitations and unresolved challenges strongly justify the proposed multi-source energy-harvesting autonomous sensor node as a meaningful contribution to this research area.

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